

Tomahawk

17315



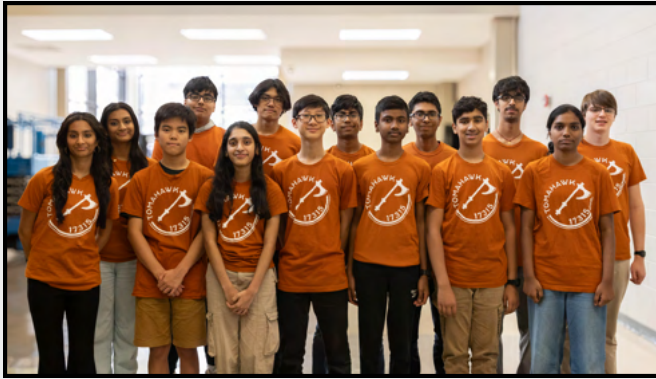
Engineering Portfolio

2025-2026





OUR TEAM



We are team #17315, **Tomahawk**, a robotics team from Westwood High School. Our team is made up of a mix of **rookies** and **veteran members**, all excited to dive into the world of robotics together. Each teammate brings **unique traits**, whether in designing, coding, building, or so much more. With fresh ideas and determination, we aim to create a robot we're proud of while also having fun with this year's *Decode* game!

Ms. Carvajal Deekshita



Head Coach

Ritvij



Captain

Aditya



Marketing
Lead

Karthik



Software
Lead

Karthik



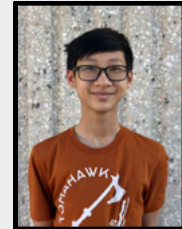
Hardware
Lead

Prahastha



Outreach
Lead

Isaac



Marketing

Aron



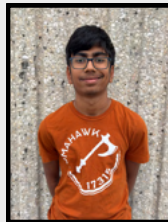
Hardware

Minessh



Software

Shrikar



Hardware

Arjun



Hardware

Sanaika



Software

Saloni



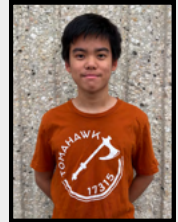
Marketing

Jusuf



Outreach

Daniel



Outreach

SUBTEAM GOALS

OUTREACH

- Build **connections** with other teams and industry professionals
- **Inspire** others, both locally and internationally to pursue robotics, and give them the tools to succeed
- Acquire **sustainable funding** for our team's robot and outreach endeavors to ensure we can **sustain** projects

DESIGN

- Make designs that are **consistent** and **long lasting**
- Implement **master sketches** for our design to better plan it out this year
- Make design decisions based on game **analysis**
- Plan an initial design and attempt to **iterate** on it unless not possible

SOFTWARE

- Create a robust, modular, and abstract **architecture** of robot functions using finite state machines
- Customize a **consistent** well scoring autonomous program
- Implement **Pinpoint accurately** to ensure the robot is able to fully **track its position** across the field.

HARDWARE

- Create an **innovative** robot using custom manufactured parts
- Implement **industry processes** and advice we learn from our **connect** meetings and facility tours
- Ensure consistency by implementing precompetition **hardware checks** to ensure robot weak spots are covered

CONTENTS

- 1: Our Team
- 2: Team Plan
- 3: Industry Meetings
- 4-5: Our Projects
- 6: Mentorship
- 7: Inspiration
- 8: Robot Overview
- 9: Engineering Process
- 10: Drivetrain
- 11: Intake
- 12: Transfer System
- 13: Outtake
- 14: Autonomous
- 15: TeleOp



TEAM PLAN

TEAM BLUEPRINT

- Ensure the **success** of **future** FIRST generations by:
 - Impacting** current FTC and FLL teams
 - Introducing FIRST to our **local community** through our **local events** to inspire them to join robotics
- Introducing FIRST to the **international community** by creating programs to give kids the opportunity to participate in Robotics
- Introducing** young minds to FIRST
- Embrace the opportunity to **learn** as engineers
- Learn and utilize **industry-standard** engineering processes
- Connect** with our community and industry mentors to further our learning and understanding
- Fundraise** money to finance our team's and club's efforts

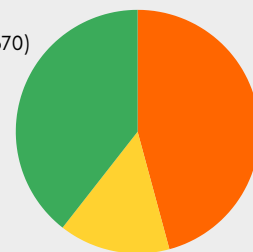
\$6670

**FUNDS RAISED
BY
TOMAHAWK**



**OUR ROBOT AT THE
CUPPRIMO SPONSOR
PITCH**

Outreach Efforts (\$2670)
39.4%



Robot (\$3100)
45.8%

Pit Design (\$1000)
14.8%

Financial allocation chart and budgeting

**For more information on our finances
please reference our Business Plan.**

OUTREACH RESULTS

Connecting with our community	8 Industry Facilities toured 38 Industry Professional Meetings
Ensuring the future of FIRST and our Team	210+ FLL mentoring hours 400+ children exposed to STEM 1 sustainability plan for our teams future implemented (see below) 3+ FTC teams mentored 2+ international regions FIRST teams created

SUSTAINABILITY PLAN

Due to the majority of our team in past years being either 11th or 12th graders, when they graduated, **our team lost a lot of expertise and knowledge last year** having to deal with many learnings simply due to not having a sustainability plan in place, **prioritizing winning at the moment over long term membership** leading to knowledge gaps on our team last year. To ensure this wouldn't occur to our current team, **we decided to recruit four 9th graders** for our team to learn from our more experienced members and from mentors, to ensure the survival of hard-earned knowledge for years to come.



Demian, our Programming Coach and current 11th grader **teaching Minessh, an FTC rookie** how to implement Pinpoint correctly

TEAM EXECUTION PLAN

ESTABLISH GOALS

- Create a **competitive** and **efficient** robot
 - Learn from **Industry Professionals** and mentors to better organize our Outreach

IMPLEMENT IDEAS

- Learned **industry standard practices** to streamline our design and build process
 - Apply the lessons to manage our Outreach projects and make smart decisions during the projects

LESSONS LEARNED

- Project management** and **industrial design** processes
 - Learned how to **connect** and **impact** the future generations of the FIRST
 - International outreach have **challenges** due to **lack of communication and distance**



CONNECT | Industry Meetings

Our plans and actions to acquire new mentors and connect with STEM industry professionals

ACQUIRED MENTORS

KEY: **ORANGE** = preexisting mentor | **BLACK** = Newly Acquired Mentor | **NAVY** = Team Alumni Mentors

Josh E.



Contribution:

-FRC 2583 alumni,
Hardware and
Strategy mentor

Rodelio C.



Contribution:

- Former Robotics club
sponsor, Philippines Outreach
guide, Fundraising guide

Santosh C.



Contribution:

-Former 19853 Roboteers coach,
Judging, Outreach, and Presentation
feedback, Design process guide

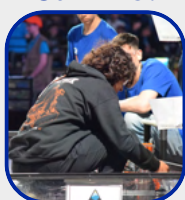
Amit U.



Contribution:

-Former 20240 coach,
Design mentor,
Manufacturing mentor

Satvik C.



Contribution:

-17315 Tomahawk
alumni, CAD
mentor, Hardware
mentor, Design
mentor

Demian M.



Contribution:

- 20716 and
19746 alumni,
Roadrunner
mentor and
debugging help

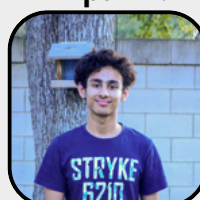
Anvitha K.



Contribution:

-18030 alumni
Helped with
presentation
and pits
preparation.

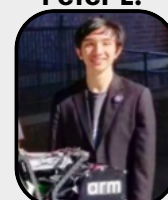
Nipun N.



Contribution:

-6210 alumni,
Awards, Software,
and Presentation
mentor, Outreach
mentor

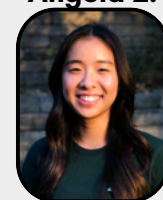
Peter L.



Contribution:

-12533 alumni,
former 19746
mentor, Networking
advice, Software
mentor, Design advice

Angela Z.



Contribution:

-4545 Ouroboros
alumni, Presentation
advisor,
Portfolio and
Presentation mentor

CONNECTING WITH PROFESSIONALS

ONLINE MEETINGS

We utilized online meetings to connect with over 40 industry professionals, learning from their practices and seeing how we can implement them ourselves. This helped us be more efficient as a team, and **also developed and improved our design and outreach process**. From these meetings we were also able to **acquire some mentors for our team including Mr. Santosh from Microsoft**.



FACILITY VISITS

Besides meeting industry professionals through online, we wanted an **immersive and engaging way** to learn about real-world processes. To achieve this, we scheduled **tours with local industry facilities** to see firsthand the practices used to improve efficiency. These **offered us deeper insights and practical lessons we could apply**.



For more information on ALL our mentors, meetings, industry tours, and their specific contributions please reference our Outreach Portfolio.

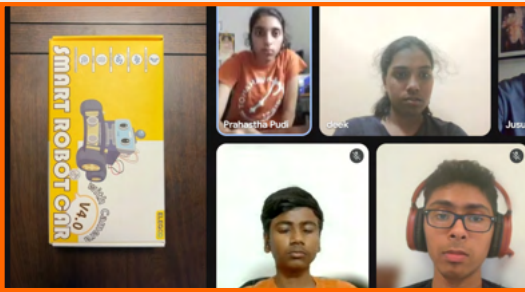


REACH | OPPORTUNITY

Giving kids the ability to participate in robotics and ensure they are able to maintain success

PROJECT PHILIPPINES

This was one of our ambitious outreach efforts this year. We found a school in the Philippines through our former mentor Mr. Creer. **We decided to send over 15 Elegoo smart kits and Spiderbots**, along with a presentation on how to build and code them. Afterward, the school's principal shared how grateful she was for the experience. Hearing this, **Mr. Creer encouraged us to consider starting the first FTC team in the Philippines**, and so we **began to research logistics** of what it would take. One of the main issues we faced was raising money. Another issue we soon found was a lack of parts in the area, realizing that we would need to send shipments to ensure the team did well.



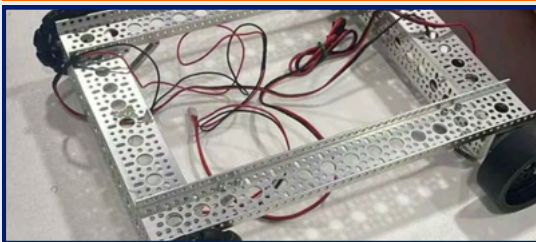
SETBACKS

Challenges we faced creating **31786 PHILBOT26 (The Philippines First FTC Team)**:

- **Registration in the FIRST in Vietnam region** (closest region to them)
- **Long shipping times** and many delays in parts arrival making setbacks in the calendar.
- Finances since the **Cebu area had just gone through a hurricane**
- The school told us it would be **difficult for them alone to pay the costs.**

Execution

We began to reach out to other potential mentors. One being **FRC team 9715**—the only FIRST team in the Philippines—and another being **faculty from the University of Cebu**. Thankfully, these two sent **mentors in person** to be able to assist them with questions that couldn't be answered online. The University of Cebu have also **lent their makerspace for Philbot**. However, we still **lacked major funds to allow them to compete at an international level**. To allow them to do so, **we taught them how to fundraise.**



COLLABORATION

Eventually, Team **11212 The Clueless** reached out to us on behalf of the **Ignite program**, founded by **11212 The Clueless and 18763 Texpan**. Ignite had prior experience developing teams in regions like South Africa and Mexico. Because they had **greater financial resources** and an **established curriculum** from similar outreach efforts, we felt they were well suited to **support more efficient teaching** for the students. Together, we were able to advocate for funds and send parts.



CONCLUSION

Despite having to face multiple issues, we still persisted in our mission. One of the biggest setbacks was part of our **part shipment being confiscated by the customs agents in the Philippines**. Another main issue we faced was having consistent meeting times due to lack of technology as well as **bad weather in the Philippines**. However, we persevered and as the current moment, due to our collaboration, the team is thriving, **working towards competing at the Vietnam Championship** as the first Philippines FTC Team.



REACH | OPPORTUNITY

Giving kids the ability to participate in robotics and ensure they are able to maintain success

PROJECT DUNE

Project Dune was one of our most successful and impactful initiatives this year. Our goal was simple: **to help establish and sustain a new FTC region in the UAE**, where interest in STEM has been rapidly growing. We learned about this opportunity after **connecting with Team 23400 UForce**, who shared that they had previously competed in the Kazakhstan region but were now working to create a Dubai-based region so teams could compete closer to home and introduce FIRST robotics to a new audience. Because the project required extensive planning and **\$30,000 in funding**, they reached out for partners. We were honored to work with **UForce, 21813 Pro2Type, and Unique World Robotics** to help make this vision possible.

Process and Result

- **28 FTC Teams** established
- To ensure **sustainability of the mission**, we mentored teams **AMD** and **Orbit** throughout the season
- Our goal was to provide mentorship to **prevent students from losing interest**, which we had identified from past experience as a major challenge.

28

**Teams
helped
created**

360

**People
impacted from
Project Dune**



Our letter of thanks and confirmation of our efforts from Mr. Bansan Thomas George, the PDP of the UAE region and the CEO of UWR.

ROBOCAMP

We volunteered at and hosted a complete student volunteer run **3-week full-day summer camp**, with over **135** unique elementary and middle school children.

- We led the children through a **series of STEM-related activities**, with the ultimate project being a **FLL/FTC robot** that each group built and programmed to play off in a **mini-tournament** at the end of the camp.
 - Our team members led the **planning** of activities, **curriculum** design, and **purchasing** of materials, and **led the camp** itself.
- Overall, our team spent **245+ volunteer hours running the camp** and acquired a strong **interest base** for our **tutoring services**.

Elementary School Week

For Elementary Schoolers, we held 2 weeks worth of programs for our local community. Most of the activities that we held were simple STEM activities such as making **DIY Lava Lamps and Pop Rockets**. However, we would also have the kids learn robotics related skills as well such as how to **build a fully functional robot out of FLL parts that with the right code would be able to follow a certain path**. We also later splits the kids into groups to make a battle bot that would compete against other groups robots to see which group **could make the strongest robot**.



**Assisting
elementary
schoolers
with STEM
related
activities**



Middle School Week

In the 1 Week program we held for Middle Schoolers, we taught them how to **make an FTC Chassis** for the main part. Our main goal from this week was to ensure they **knew how to build and program a fully functional drivetrain** as well as to give them a taste of FTC.



245+

**VOLUNTEER
HOURS FROM
TOMAHAWK**

135+

**ROBOCAMP
ATTENDEES**

Teaching Middle Schoolers how to build(and code) an FTC Chassis and showing them how our own robot looks



REACH | MENTORSHIP

Impacting and guiding our local robotics community to higher levels.

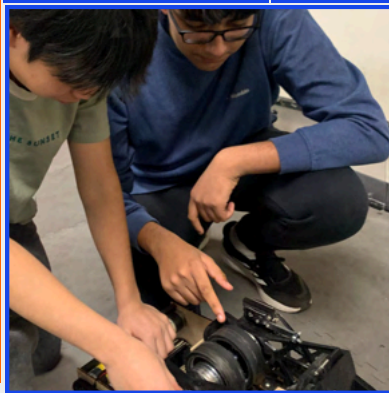
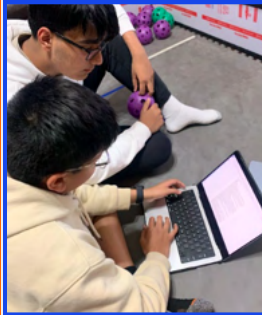
OUR MISSION

To lower the barrier of entry for robotics and STEM, making robotics and success more accessible for all students.

This year, through our meetings with professionals and our interactions with our local community and abroad, we wanted **one common goal and theme for this years outreach**; a final product and **lasting impact** that all of this work would help create. Together, from our early year connect meetings and some of our **past experiences** on other teams, we set a new goal that we hoped to achieve this year through all of our efforts. We chose this specific goal because of our **past experiences**, that when we all started FIRST, **many of us discovered robotics very late** and also from our experience, we realized even then, **we didn't have much help or resources to help us learn fast.**

TEAM MENTORING

This year, we also attempted to mentor many teams to ensure the **sustainability of their teams**. From our past experiences, we had learned that many times people would quit **because of a lack of support** which would demotivate them and cause them to not want to continue to participate. We wanted to **ensure continued interest** because we believed that every team had the ability to do well or have a great passion for robotics but **they may not have the right guidance or advice** to do so. Thus, we reached out to teams offering our mentorship.



TEAMS WE'VE ASSISTED

18886 Atlatl - Robot design, Portfolio help, **season mentorship**

17113 Hunga Munga - Robot construction, software assistance, CAD lessons, and debugging help, Outreach Help, **season mentorship**

28588 AMD - Robot construction help, software assistance, **season mentorship**,

30358 The Gearchaeologists - Robot Stability, consistency, **season mentorship**, leadership, portfolio help, marketing help

24167 Orion - drive coaching help, strategy help, robot design help, **season mentorship**

11503 Hyperfang - Robot Design advice, drive coaching help

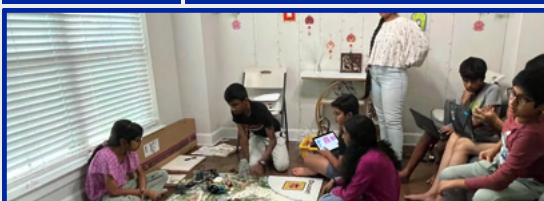
28831 Orbit - Robot Design Advice, Software help, **season mentorship**

32610 Multiversal - Hardware help, Robot Design Advice

31786 PHILBOT26 - **Team creation**, mentor finding, hardware help, software help, debugging help, portfolio help, **seasonal mentorship**

FLL MENTORING

Over the course of the season, we would mentor FLL team 66670 Spicewood Superstars and team 73061 Spartabots on a **weekly** basis assisting them with **executing their missions** efficiently and **preparing for FLL judging interviews.**



"Tomahawk has been invaluable in this process- discussing ideas with the kids, providing thoughtful feedback, and helping them reach those 'aha' moments.

They then played a key role in getting the team's innovation project presentation ready, from structure and clarity to confident delivery. With tips on body language, pacing, and audience engagement, Tomahawk helped turn the kids' nervousness into excitement, leaving them confident and ready for the judges and audience."

-Prashant Baronia,

FLL team Spicewood Superstars 66670 coach

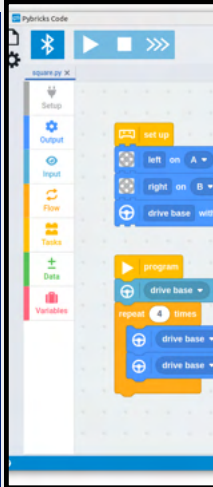


REACH | INSPIRATION

Giving kids the inspiration to get into robotics and STEM

FLL TUTORING

In order to spread STEM to the younger generation, we decided to **offer FLL tutoring services, teaching individual kids for to design and code competitive FLL robots so many kids could get the opportunity to learn** even if they didn't have the resources. This idea was inspired from a Robocamp camper named **Bharakh** who reached out to us at the conclusion of Robocamp asking if he could learn more materials similar to what he had learned at the camp. From him we realized **we could tutor kids who have a passion for robotics**. Our tutoring had a specific curriculum consisting of **FLL Robot assembly, PyBlocks, and Python lessons**.



Bharakh, one of the kids that we tutored after being inspired from robocamp, building an FLL bot and coding his robot in blocks, mentored by Daniel.



Veda is another kid tutored by Sanaika and Saloni. She is learning **basics** of robotics by exploring the **SPIKE essentials kit**.



COMMUNITY EVENTS

This year we also wanted to raise awareness about our team in **our community** too. These events not only allowed us to reach more community members through different interest groups, but also **allowed for us to raise awareness about FIRST** and our own initiatives such as our FLL tutoring and our projects very easily. These Community Event displays also allowed us to learn how to better raise interest in younger kids for robotics and **we could use that learning to better teach the teams we mentored and the kids we tutored**.

For more details of ALL our community displays, please reference our Outreach Portfolio. (at pits)

Anderson Mill Carnival

We went to the Anderson Mill Carnival held by the Anderson Mill PTA. Here, we let the kids **drive the robot** and explain to them a bit about our team as well as our teams robot from the Into the Deep season. We also used the festival to help tell parents about our **tutoring services** we offered.

Back to School Bash

At this event almost 1,000 people came in total including many RRISD board members. Here, we gave a robot demonstration to the kids who came and allowed them to drive it. We also spoke with RRISD board members to explain the benefits robotics and STEM had on us.

Diwali in The Domain

At this event, we not only told people about ourselves but we also were able to **reach out to local businesses also at the event** and be able to get their contact to potentially connect with them. We also were able to **connect with potential industry mentors** who were interested in helping.



Giving kids a robot chassis to drive around at the Anderson Mill Carnival



Talking with RRISD Educators about the impact of robotics in our lives



Kids at the Diwali in the Domain festival driving around a robot

Overall Impact

1600+



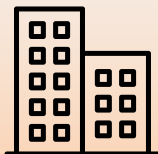
People impacted

30



Meetings with professionals

8



Industry Standard Facilities Toured

10



Sponsors acquired

12



Mentors gained

10

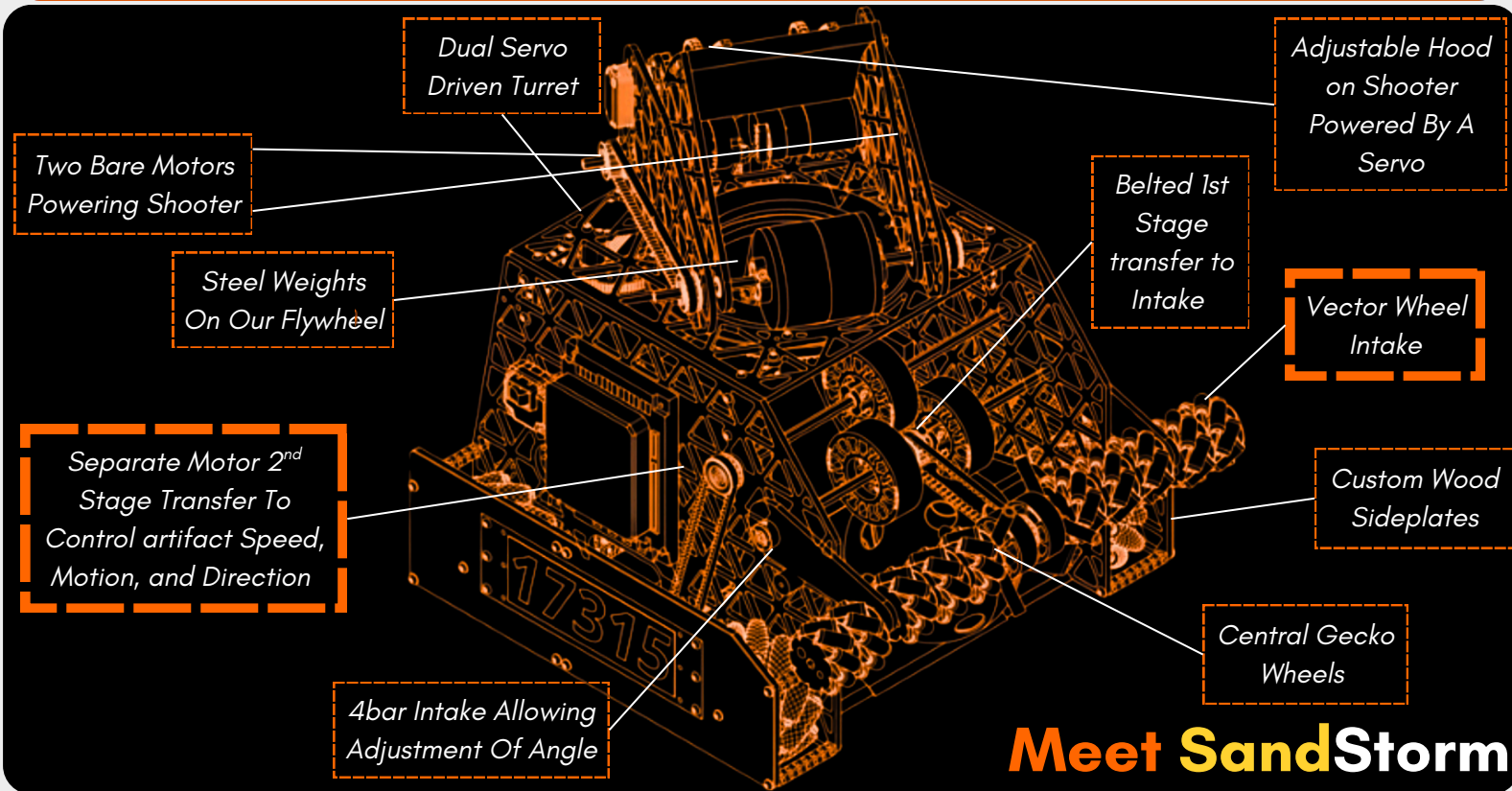


Total teams mentored



DESIGN | ROBOT OVERVIEW

This year we wanted to create a robot that would score well from **both the Large Launching Zone and the Small Launching Zone**. One more realization we had from analysis was the true **value of Ranking Points**. Thus, we wanted to create a robot that could **consistently be able to get 5 or 6 ranking points** every game **by ourselves**. This would ensure we would **rank higher** and allow us to be in a better position going into the elimination rounds.



GAME STRATEGY

We have two distinct strategies that we have developed to help us score more efficiently during games. **Our Far side strategy** allows us to average **180** points per game solo while our **Short side strategy** scores **150**.

Far Side

Autonomous Period:

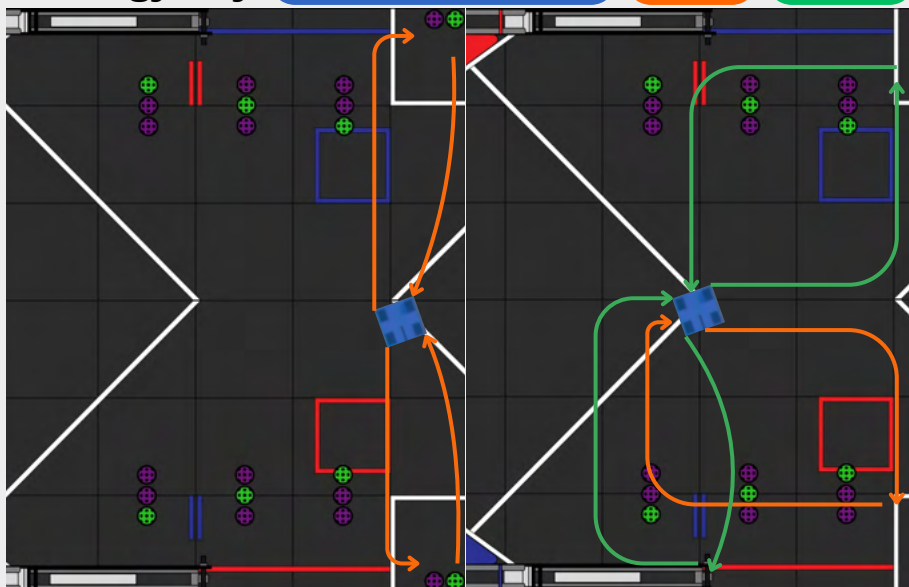
- Shoot **preloaded** artifacts.
- If with **alliance** who **clears gate in auto**, grab artifacts rolling into the human player zone. Repeat this twice.

TeleOp Period:

- Initial Move: Let **alliance** clear gate
- Core Cycles: Retrieve artifacts primarily from our **own Human Player Station** or from **opponents Human Player zone** depending which one is safer and has more artifacts.

Strategy Key:

Autonomous starting position 1st Instinct 2nd Instinct



Far Side Shooting Strategy

Short Side Shooting Strategy

Short Side

Autonomous Period:

- Shoot **preloaded** artifacts.
- Collect and shoot all nine additional artifacts.
- End the period positioned in the large shooting zone

TeleOp Period:

- Initial Move: **Clear gate** first.
- Core Cycles: Retrieve artifacts primarily from **our own Human Player Station** or **directly from the gate**.
- Opportunistic Plays:** Use the opponent's station if it's unguarded and accessible.



THINK | ENGINEERING

DESIGN PRINCIPLES

Our robot is designed to be **modular** and **maintainable**, aligning with our goal of simple yet effective.

M * 2

We used MAYA, which stands for **Most Advanced Yet Acceptable**, to help us balance creating a **consistent robot** with creating a **high scoring robot**

MAYA

We avoid unnecessary complexity by **Keeping It Simple, Stupid** to focus on robustness and efficiency of the design.

KISS

We implemented **SPARC**, a **customer based design process** that caters to the desires of consumers. We implemented this by catering our robot design to be easier for our software team and for our drivers.

SPARC

We avoid **premature optimization**, first ensuring reliability and consistency. Make it work, then make it fast.

P.O.

We used **RAID**(Risk, Assumptions, Issues, and Dependencies) in our planning, we hoped by **identifying our own strengths and weaknesses**, we could be realistic in our goals and complete them more often.

RAID

For more details of how we learned these techniques, please reference our Hardware Portfolio. (at pits)

DESIGN PROCESS

Using our **learnings** from some of our Outreach Connect Meetings, we decided to create our **seasons design process** that we wanted to follow that could combine the best of all worlds from all of these design principals that we had learned to ideally create an **optimal design process** which would ensure our bot was **unique, modular, and consistent**.

Plan → **Design** → **Simulate** → **Build** → **Test**

Inspired from our Tesla tour, Microsoft tour, our Indeed building visit, and our meeting with Mrs. Rajendran, we knew that planning would be essential. Using **Tesla's SPARC mindset**, we attempted to plan around **our drivers' and coders' skills** and desires similar to how Tesla planned around their **customers**. We also **defined** our **constraints** from the start of the time and resources we had available. We also wanted to plan better to avoid **the Pareto Principle**. Inspired by **Microsoft** we also used **NUDDs; New, Unique, Different, Difficult** ideas as a core part of our planning phase.

From our meetings, we were inspired by Tesla's **MAYA principles** and applied them to our design process. We aimed to incorporate scoring features that distinguished our robot while ensuring simplicity and consistency in performance. This also aligned with the principle we learned in our meeting with Mr. Hwang's **KISS principle**. Another principle we hoped to implement was **complete CADding**. Last year we had to make **many adjustments because we didn't cad very detailed**. This year our hope is to be able to avoid that by **cadding everything, down to every last screw**.

Our next step was to simulate every material we use as well as prototype our designs with materials we had. Not only to create all of our components in a Computer Automated Design software(CAD) but to also **simulate stress and tension** where it was applicable. We attempted to design our parts to be sturdy and to last. Our team policy was to **stress test everything that was made of wood or aluminum**, as those were the components that were the most susceptible to damage. This was **inspired from our teams tour of Tyrex and TMI** where we saw their usage of stress test machines.

The next step we had in our design process was the actual build phase. this phase was designed to assemble together the components that we designed. Our main goal from this phase was to assemble the robot in such a way that it wouldn't break. To achieve this, we used **loctite** and other thread lockers to ensure the screws would be secure. Also in this step, we decide to have a **hardware check** the day before our competition to identify any potential risks on our robot and correct them as soon as we could. This precaution we hoped would allow for **more consistency in performance**.

The final step of our design process is testing, which plays a critical role in improving both our designs and overall robot performance. **Inspired by Tesla's review-based design process**, we continuously evaluate our mechanisms during and after production. Because components are almost always in use on the robot, testing is ongoing rather than a single step. **Following the principles of Kaizen**, we regularly identify areas for improvement and iterate by **returning to the design or build phase** whenever performance falls below our standards.

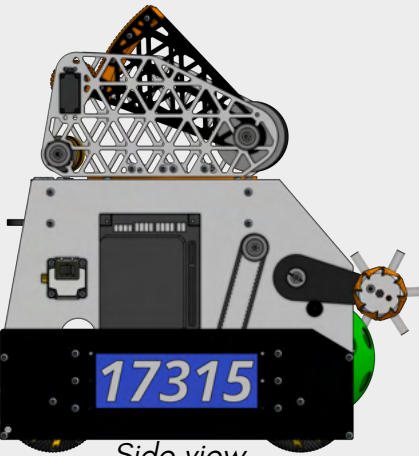


DRIVETRAIN

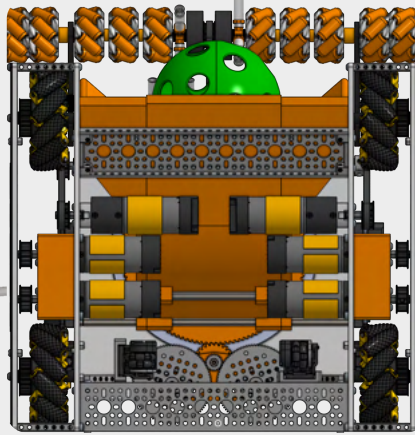
DESIGN GOALS & CONSTRAINTS

When designing our drivetrain, we wanted:

- **Low clearance** to improve stability and **lower the Center of Gravity**
- Be **modular** to allow for easy iterating
- Provide **uniform wheel movement**, allowing for more **accurate** programming.



Side view

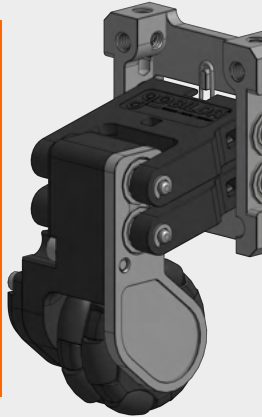


Bottom view

ODOMETRY MODULES

We use a version of **four-bar** odometry, developed by goBILDA for easy integration with their parts.

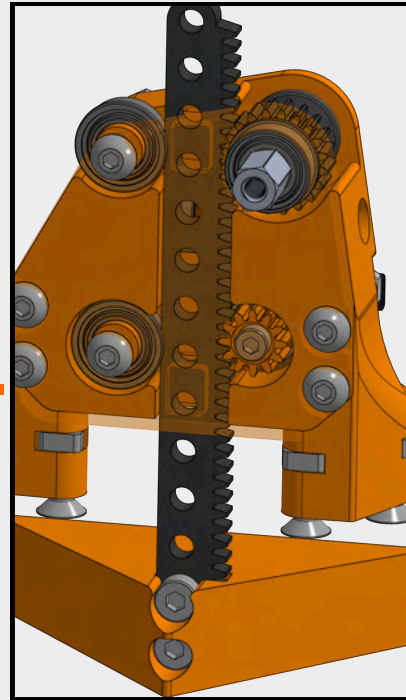
- **Low profile** four-bar sprung localization modules
- Built into cross robot U-channels, allowing for:
 - Spatial **efficiency**
 - **Accurate** tracking at high speeds



OVERVIEW

Our drivetrain features a **parallel plate** design that uses 4 mecanum wheels, allowing for quick **maneuverability**. The main structure is composed of 4 laser-cut wooden plates, which can be easily removed with **6 screws**. This design enables us to quickly **iterate** and change our subsystems rapidly such as **our intake** and **our transfer geometry**.

Future Implementation / PARK/CLIMB



Our Climb design made up of a servo driving a rack and pinion mechanism

This year, we also wanted a plan to be **able to get 2 full parks consistently**. Initially, we thought that creating a design would be difficult and would require the **use of Misumi slides and two motors**.

However, we soon realized that we that **we could design a park mechanism that would elevate the robot only a few millimeters off the ground** and that if done in **joint with another team, we could both reach 2 full parks**.

However, due to a **lack of time, we decided to postpone** this design to be implemented for later competitions such as areas or worlds.

MAINTAINABILITY THROUGH ACCESSIBILITY

The **modular** design of our robot enables **quick iteration** and repair of subsystems. Between League Meets, we've **completely redesigned our intake and outtake mechanisms**, but our maintainable structure allows us to modify and replace the old system with a **maximum of 8 screws**.



Outtake - **8** screws to detach

Intake - **2** screws to detach

Transfer Stages 1&2 - **4** screws to detach

Access to Drive Pod - **6** screws

P Plan: Design a stable, **well built chassis** that is constructed in such a way that it is easy for our software team to tune well and is **easy to package all our mechanisms** allowing for **greater customizability and creativity**.

D Design: Make a Parallel Plate Drivetrain (PPDT) which allows us to have our **motors belted to our wheels**. This gives us much more freedom in the placement of our motors helping us package our mechanisms better.

S Simulate: Review our CAD to **identify any mistakes** or potential issues, such as **inaccessible areas for wiring or fasteners**. Run tests such as **center of gravity (CoG) analysis in Onshape** to ensure our robot will remain stable. Also, we used a **6 mm wooden chassis**, thicker than the typical 4 mm, to **better withstand the high-contact** nature of the game.

B Build: Buy the necessary parts to assemble the chassis and begin to combine the parts together to make the chassis.

T Test: Evaluate how well the chassis works and **make adjustments based on the data we collect** (Ex: we noticed that our wheels had a great amount of resistance when we spun them and **added washers on our axles** that we connected to the plate, **to allow the wheels to spin more smoothly**).



INTAKE

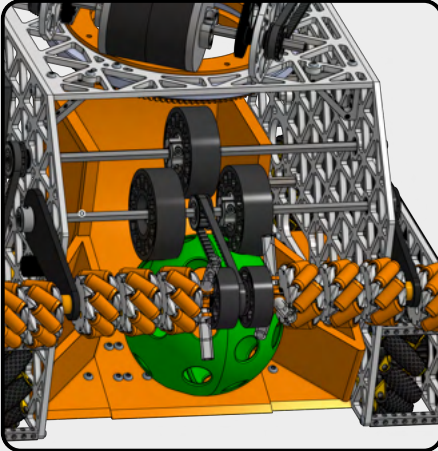
INTAKE V3: Vectored Intake + Tubing

DESIGN GOALS & CONSTRAINTS

We wanted an *intake* that could:

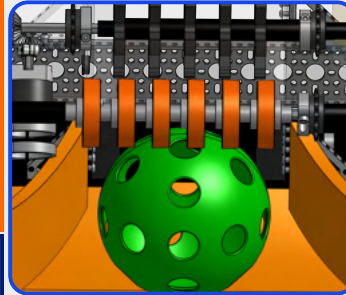
- Intake from the **corners** of the Human Player zone **efficiently**
- Intake quickly from the **ground**
- Intake samples **reliably** and feeds to **transfer**

This made our robot more **consistent**, allowing for a more reliable autonomous and **faster** cycles during teleop.



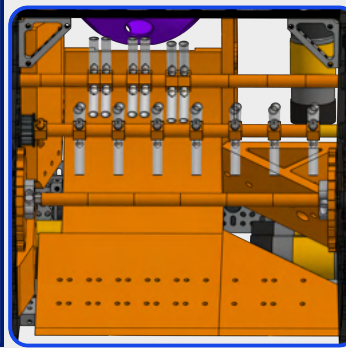
In our current iteration, we use **custom made vector wheels** whose unique slanted shape **allows them to pull artifacts towards the center** where we have two geckos that pull it in. We also switched to back to gecko wheels due to tubing being inconsistent. **This switch helps prevent potential dead spots** in our intake ramp and transfer, an issue we faced with our V2.

V1.75: *Gecko Wheels*



In our first main intake iteration, we used **Andymark gecko wheels** and **Gobilda boot wheels to intake**. We also had a curved ramp geometry which we felt would allow the artifacts to travel faster. **We Learned** that the **ramp was too curved** and that gecko wheels alone were slow **due to being smaller** and not having as much "pull" on the artifacts.

V2: *Rubber Band Intake + Tubing*



In our second main intake iteration, we redesigned the intake by **switching to a rubber band intake**, using tubing instead of a boot wheel, and **straightening the ramp**. **We Learned** however, that the **rubber bands broke easily** and the **ramp geometry created dead spots** where artifacts stalled before reaching the tubing.

For more details on our robot mechanism iterations please reference our Hardware Portfolio. (at pits)

- P Plan:** Brainstorm an intake that allows for **artifacts to be pulled in almost instantly**. Also, to create the geometry in such a way that there are **no dead zones for artifacts to jam in** or **areas the artifacts cannot move past fast**.
- D Design:** Create **sketches for the intake** and start to **design it in Onshape**. Use sketches to create **geometries of the ramp** and use **measurements** to know where to place any **hard stops or bearing holes** for the intake.
- S Simulate:** Test different materials and evaluate which ones would be the best. Some ways we implemented this for our intake was through the **type of rubber bands**(we used thicker ones) and types of materials for the arm(we used **1/4" polycarbonate** since it was **easy to find**, relatively **easy to manufacture**, and was **cheaper** than aluminum.
- B Build:** Buy the necessary parts to assemble the chassis and begin to combine the parts together to make the chassis.
- T Test:** Evaluate how well the chassis works and **make adjustments based on the data we collect**(Ex: we noticed that our wheels had a great amount of resistance when we spun them and **added washers on our axles** that we connected to the plate, **to allow the wheels to spin more smoothly**).

INNOVATE | VECTOR WHEELS COVERS

One issue that we originally faced with our intake was a **lack of grip on the vector wheels**. We decided to fix this, we decided to **add tubing covers onto the vector wheels** to give more grip. By testing different types of tubing and researching, **we found that silicone tubing worked best due to its durability** compared to the traditional latex tubing.



THINK | 4-BAR HARD STOP PLACEMENT

When we originally designed the 4-Bar hard stop, **we didn't account for tile sink**(since it is very hard to tell from simply Onshape) or **accurate artifact compression**. Thus, we had to adjust the hardstop position. There were two ways to do this, by **increasing the radius of the hardstop**, and by **changing the position of the actual hole**. Through extensive testing, we kept adjusting these two variables until we finally found a **combination that allowed us to have good compression and grip on the artifacts**.





TRANSFER SYSTEM

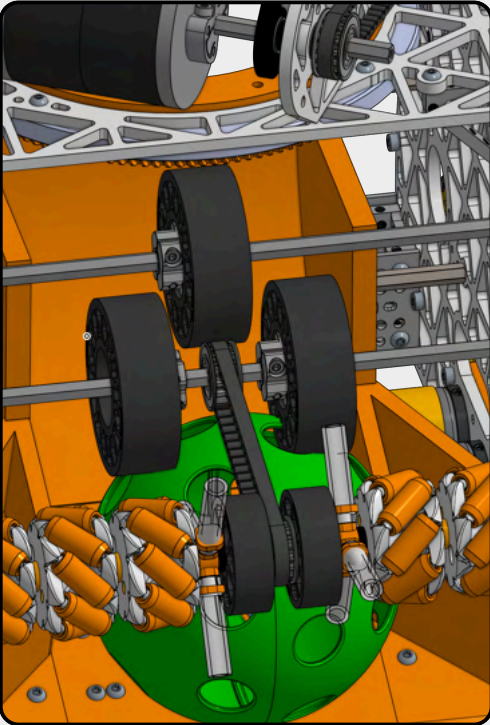
For more details on our robot mechanism iterations please reference our Hardware Portfolio. (at pits)

TRANSFER V4: 2 Stage Surgical Tubing

DESIGN GOALS & CONSTRAINTS

We wanted a transfer that could:

- Transfer **rapidly** from the Intake to the Outtake
- Was able to **clear the upward slope** up to the outtake
- Would work **consistently** (an issue we faced earlier in the season)



In the fourth iteration, we **changed to a vector-wheel intake** that actively centered artifacts, **removing the need for a left-sided transfer** that had required extra guides and **caused instability**. This allowed us to **redesign the transfer as a centered, symmetric system**. We also switched from **Silicone tubing** to **gecko wheels** as although tubing had worked, its over **usage made it needed to be replaced** often which was **costly and time draining**.

P Plan: Think of a way to transfer the artifacts in such a way that it **doesn't affect cycle times negatively** and allows us to **control when we want to shoot and when we want to hold balls**.

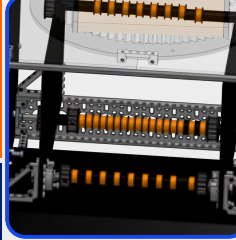
D Design: Start **creating geometries** in Onshape of our outtake and our intake and start to calculate how we can make the geometry so that it **transfers without jamming** or **slowing down**.

S Simulate: Start **modeling how our transfer** would look, using **prototypes to gather information** about how **effective** using **certain materials and structures** would be. Ex: testing rubber bands for our V1 Conveyor mechanism

B Build: Putting together our CAD and making each part out of the **materials we experimented with** and found were good for the purpose, in the case of the transfer, **the cost benefit analysis for using Geckos vs. Surgical Tubing**.

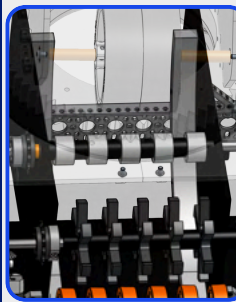
T Test: Testing our transfer, and making changes where needed. One of these changes we made was **strengthening our surgical tubing 3D printed cores** due to them **snapping a lot under pressure from the artifacts**.

V1: Rubber Band Stages



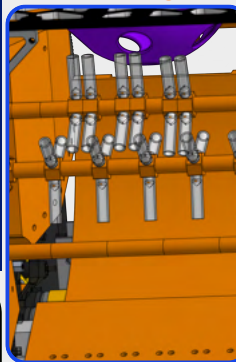
This was our first iteration of our transfer. This transfer was made with the idea to act like a **conveyor belt for the artifacts**. However, it quickly **failed** after we realized it **wouldn't be feasible** due to the possibility of **artifacts simply slipping off the conveyer**.

V2: Rails + TPU rollers



This design consisted of **"rail" like guides** for the artifact to travel on with TPU rollers to help push the artifact up into the outtake. **We learned** from this however, that **the rollers were inconsistent** and the geometry relative to the turret made it very unpredictable. The TPU rollers also **slipped** often and were **too small to have enough grip** to rapidly transfer the artifacts.

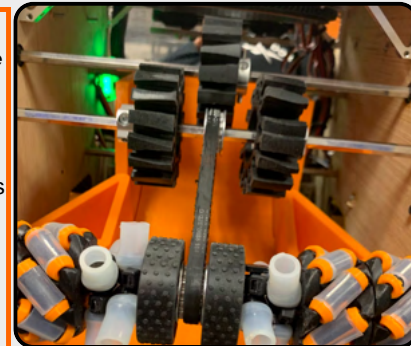
V3: 2 Stage Surgical Tubing w/ guides



Due to the new intake, we redesigned the transfer to the left side, **adding guides to direct artifacts straight to the outtake**. We also **reduced the outtake hole size to better match the artifact's diameter**, ensuring consistent placement. **We learned** from this iteration however, that the ramp was **unstable due to the transfer guides being separated from the main transfer geometry**.

INNOVATE | Cut Gecko Wheels

When we were designing our transfer, one issue that came up was that the **gecko wheels** had a much greater radius **compared to tubing** which led it to move the balls slower despite it being sturdier and more secure than tubing. **To account for this disadvantage, we made cuts in our 72mm gecko wheels to allow us to process artifacts faster**. By cutting the gecko wheels, we **got tubing like geometry to pull the artifacts better** while being sturdier compared to tubing leading to better intaking efficiency.





Outtake

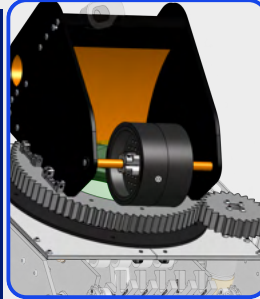
OUTTAKE V3: Adjustable Hood + Weights Flywheel

DESIGN GOALS & CONSTRAINTS V1 - Base Shooter

We wanted an *outtake* that could:

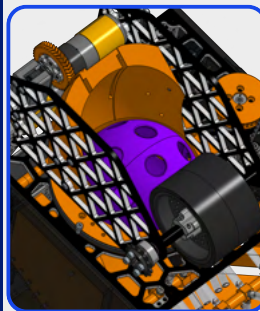
- Be able to adjust to different positions on the field and still be able to shoot accurately to make it difficult to be defended.

In our final outtake iteration, we used a **dual-servo turret** and a **separate hood** to **improve stability** and **smoothness** over the **previous bearing design**. We added **custom 72 mm steel flywheel weights** to increase rotational inertia and improve recovery time, allowing us to **rapid fire artifacts** together. We also switched to an **AndyMark turret bearing** for greater **consistency** than with the **bearing stack** and easier **packaging**.



Our original shooter, consisting of two **96 mm Rhino Wheels** and a **fixed hood**. As our first shooter, the design had several flaws. **We used a bearing-stack turret with 3D printed gears** for 360-degree rotation. **We learned** that the artifacts were **under-compressed by 5 mm**, requiring **foam to compensate**, and we learned that **better turret planning** was needed to increase the **base height** the shooter as the turret gears were too **tall** and had **immense backlash** from their **spur gear pattern**.

V2 - Shooter with adjustable hood and bare motors



We then changed our outtake geometry to have an **adjustable hood run by a servo and bearings**. This allowed us to **control the angle at which we shot at**, allowing us to **adjust it based on how close or far we were to the goals**. We attempted once again to implement turret but failed due to a lack of time. **We learned** how to manage our time better and how to better design an outtake to work well.

P Plan: Create an **adjustable outtake that is able to change the angle it shoots at**. Also to allow our **outtake to rotate 360 degrees** to allow us to **lock onto the goal** whenever we want to shoot.

D Design: Start designing the **geometries** and **calculating the angles we need to shoot at** in order to **minimize backspin** and **maximize accuracy**.

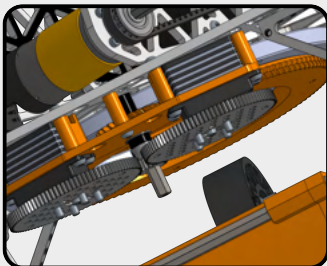
S Simulate: Test the **efficiency of different materials**. For the outtake, we tested if **steel or aluminum weights** would be ideal and **how many weights** would lead to the **greatest reduction in backspin** and **smallest increase in spinup time**.

B Build: Assemble together the robot and make **small adjustments** such as **adding foam if compression was too low**.

Test: Evaluate how well the mechanism works and see if we need to make changes. Our ideology this whole season, inspired by **Todd and Peter** was to find the part that was **slowing our cycle times the most** and **see if it was such an urgent need that we needed to fix** that would **otherwise hinder software teams ability to make it work well**.

Dual Servo Turret

As part of our robot, we implemented a **dual servo turret**, very different from our earlier motor powered turret, by using the 2 servos (**Axon Maxes**), we were able to **compensate** and use that motor elsewhere. By using **analog wires** and a **JST Board**, we were able to **almost perfectly replicate motor encoder** values.



During the season, we wanted to **implement flywheel weights** into our shooter. however, we realized although it would **increase our rotational inertia** and decrease backspin, it would **also increase our shooter spin up time**.

TRADE-OFF Analysis: Rot. Inertia vs. Spinup

Flywheel Calculations

Disk Rotational Inertia Formula: $I = \frac{1}{2}MR^2$

I : Rotational Inertia M : mass of disk (kilograms)
 R : Radius of disk (radius)

Rotational Inertia without weights

Mass: $149\text{grams} \cdot 2 + 2.1\text{grams} \cdot 8 + 85\text{g} \cdot 2$
 Bone Hubs Servos 72mm Rhino wheels
 $= 214.8\text{grams}$

Radius: 72mm $72\text{mm} \cdot 0.001\text{m} = 0.072\text{m}$

$I = \frac{1}{2}MR^2 = \frac{1}{2}(0.2148)(0.072)^2 = 0.0005567616 \text{ Kg}\cdot\text{m}^2$

Rotational Inertia WITH weights

Mass: $214.8\text{grams} + 136\text{grams} \cdot 2 = 686.8\text{grams}$ $\leftarrow$ Steel weights $\leftarrow$ 100grams
 $= 0.6868 \text{ Kg}$

$I = \frac{1}{2}MR^2 = \frac{1}{2}(0.6868)(0.072)^2 = 0.00179045568 \text{ Kg}\cdot\text{m}^2$

Percentage Formula: $\frac{\text{Final} - \text{Initial}}{\text{Initial}} \cdot 100 = \frac{0.00179045568 - 0.0005567616}{0.0005567616} \cdot 100 = 220.8\%$

Doing the math, we found that we would get almost a **175% increase on rotational inertia/ rapid fire (ability to shoot balls together)** using 2 Steel weights while only losing about **7 seconds of spinup time** (since the total spinup is much greater than what we need, the spinup time is **less important** compared to being able to have greater rotational inertia which allowed for greater accuracy).



CONTROL | AUTONOMOUS

GoBilda Pinpoint

Why This Matters:

- Removes **drift**
- Makes paths **repeatable**
- Allows **math-based** aiming
- Makes **tuning** easier
- Works even when field **friction** change

How We Use Odometry in Auto:

- We use the robot's **position** from Pinpoint to calculate:
- **Distance** to the **target**
- Angle the turret must **rotate**
- **Hood angle** needed for the shot



Pinpoint Odometry Computer

Our autonomous system is built around the **GoBilda Pinpoint Odometry Computer**, which gives us real-time **X/Y position** and **heading**. Pinpoint uses **three** dead wheels to **measure** movement, allowing the robot to know exactly where it is on the field at all times.

Math Behind It

The robot uses **odometry** to compute the **distance** and **angle** to the **target**.

This gives us **consistent** aiming without needing a camera during **autonomous**.

Hood Angle / Flywheel Velocity

We use distance **d** to choose:

- **hood angle**
- **flywheel velocity**

Robot and Target Positions

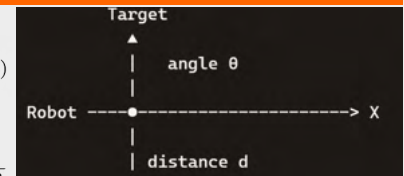
- Robot odometry position: (x_r, y_r)
- Target field position: (x_t, y_t)

Distance Calculation

$$d = \sqrt{(x_t - x_r)^2 + (y_t - y_r)^2}$$

Turret Angle Calculation

$$\theta = \text{atan2}(y_t - y_r, x_t - x_r)$$



This graph shows how the **distance** is calculated through **PID** from the starting point.

PID-Stabilized Flywheel And Turret

PID is one of the most **important** parts of our autonomous system. We use PID on both the **flywheel** and the **turret**.

Why PID Matters:

PID allows the **motors** to:

- **accelerate** smoothly
- reach a target **setpoint**
- flatten out and hold **speed**
- stop **cleanly** without overshoot
- stay **consistent** even when battery voltage drops

This is **critical** for autonomous because every shot must be stable.

Turret PID Behavior:

- **Smooth** rotation
- No jerky **movement**
- No **overshoot** past the **target** angle
- **Holds angle** even while the robot **moves**
- This **makes** turret **alignment** extremely **reliable**.
- This also makes the robot **less susceptible** to **defense**



Flywheel PID Behavior

- P** term **accelerates** the flywheel quickly
- I** term **corrects** long-term **drift**
- D** term prevents **overshoot**

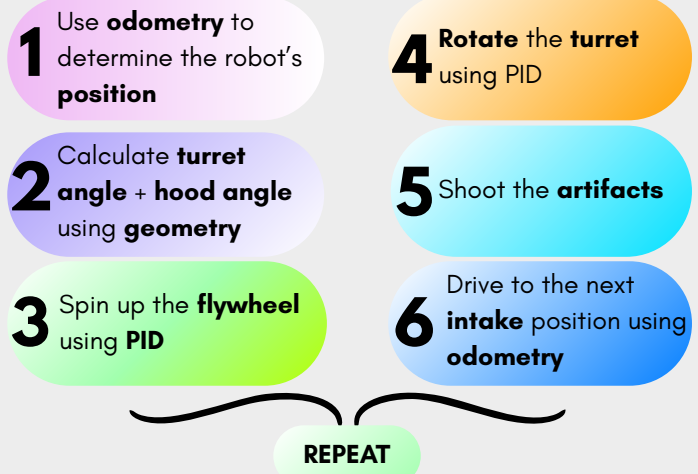
Result: the flywheel reaches the target RPM and stays flat

This graph shows how **PID** brings the flywheel to the **target RPM** and keeps it **stable**.

Without PID, the RPM would **bounce**, causing **inconsistent** shots.

How These Systems Work Together in Auto

Each cycle of our autonomous routine follows the **same** structure. Because every step is **math-driven** and **PID-stabilized**, the robot behaves the same way every time, even on **different fields**. The odometry gives us a **reliable position**, the geometry tells the **turret** and **hood** exactly where to **aim**, and the PID loops make sure the motors actually **reach** and **hold** the values we **calculated**. This keeps the whole routine **predictable**, so even if the carpet is **worn down** or the field has **slightly different friction**, the robot still drives to the **same spots** and **shoots** with the same **power**. It basically removes the **"randomness"** from autonomous and makes the robot act the way we **expect** every run.





CONTROL | TELEOP

Auto-Velocity Flywheel Control

Our flywheel uses a **PID-based** automatic **velocity control system** during **TeleOp**. Instead of the driver **manually** adjusting **motor power**, the driver presses a **single** button to spin the **flywheel** up to a preset **target RPM**. The PID controller **continuously** adjusts the motor **output** to maintain that **speed**, compensating for factors like **load** changes and **battery** voltage, which results in more **consistent** and **reliable** flywheel **performance**.

Why It Matters

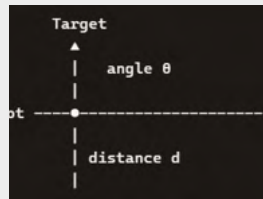
- **Consistent** shot power
- Faster **recovery** between shots
- Less driver **workload**
- More **accurate** scoring

Diagram – Flywheel PID Behavior



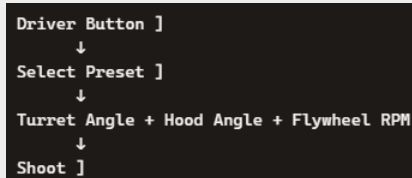
The flywheel accelerates **smoothly** up to the **target RPM** without **overshooting** and then maintains a **steady, flat speed**. The PID controller **continuously** adjusts motor output to **correct** small **disturbances**, keeping the RPM **stable** even when **battery** voltage **drops** or the load on the **flywheel changes**.

Diagram – Turret Aiming



We use the robot's **odometry** to determine its **position** on the field and **calculate** the **distance d** and **angle θ** to the target. The turret automatically rotates to the required **angle θ** , while the **hood position** and **flywheel speed** are adjusted based on the calculated distance d to ensure **accurate** shots.

Diagram - Preset Buttons



A single **button** press **activates** a **preset** that automatically sets the **turret angle**, **hood angle**, and **flywheel RPM**, simplifying operation and improving **consistency**.

Driver Presets & Controls

We added a few simple presets to make TeleOp faster and easier for the drivers:

- **Shot presets for different distances** (One button sets the **turret angle**, **hood angle**, and **flywheel speed** so the driver **doesn't** have to adjust everything **manually**.)
- **Hood angle presets** (These make sure the hood is always at the **right angle** based on how **far** we are from the **target**.)
- **Manual turret override** (If the **auto-turret** isn't lined up perfectly, the driver can take **control** and make quick **adjustments**.)
- **Optimized intake controls** (Clean, simple **intake buttons** help the drivers **cycle faster** without thinking about **timing**.)

These aren't **huge** systems, but they help the drivers cycle **faster** and stay **consistent**.

TeleOp Strategy

Most of TeleOp is **driver strategy**, not **software**. Our drivers focus on:

- **cycle paths**
- **communication**
- **timing shots**
- **avoiding traffic**

The software systems (**auto-velocity** + **auto-turret**) are designed to support the drivers so they can play **faster** and more **consistently**.



Auto-Turret Alignment

The turret automatically rotates to the **correct** angle using **odometry** + **geometry**. The driver only needs to get the robot near the **shooting** area, the turret handles the **fine alignment**.

Why It Matters:

- **Faster** cycle times
- More **accurate** shots
- Consistent **alignment** even while driving
- **Consistency** in scoring
- More **room** to think about **other** things
- Less driver **stress**